Xinyang Liu

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Research

My primary research goal is to solve real-world problems through advanced Generative AI systems capable of understanding, generating and reasoning with high-dimensional data across diverse modalities. With this goal in mind, I am working on **Generative Modeling**, including its theoretical exploration and various applications in data generation and multimodal learning

My recent research focuses on how to push a model's capability boundary beyond its own limits through distillation techniques.

Education

The University of Texas at Austin — Austin, United States
Ph.D., Statistics and Data Sciences — Aug 2025 -
Advisor: Mingyuan Zhou

Xidian University — Xi'an, China
M.S., Information and Telecommunication Engineering — Sep 2021 - Jul 2024
Advisor: Bo Chen

Xidian University — Xi'an, China
B.S., Electronic Information Engineering — Sep 2017 - Jul 2021

Publications (* denotes equal contribution)

Preprint

[1] Tiansheng Wen, Yifei Wang, Aosong Feng, Long Ma, **Xinyang Liu**, Yifan Wang, Lixuan Guo, Bo Chen, Stefanie Jegelka, Chenyu You
Route Experts by Sequence, not by Token
ArXiv:2511.06494 **(2025)**

[2] Chaojie Wang*, **Xinyang Liu***, Dongsheng Wang, Hao Zhang, Bo Chen, and Mingyuan Zhou
Scalable Weibull Graph Attention Autoencoder for Modeling Document Relational Networks
ArXiv 2410.09696 **(2024)**

Conference and Journal Publications

[3] Xinyue Hu, Zhibin Duan, **Xinyang Liu**, Yuxin Li, Bo Chen, and Mingyuan Zhou
Disentangled Generative Graph Representation Learning
IEEE Transactions on Neural Networks and Learning Systems **(IEEE TNNLS)**

[4] Haoyang Zheng, **Xinyang Liu**, Xiangrui Kong, Nan Jiang, Zheyuan Hu, Weijian Luo, Wei Deng, and Guang Lin
Ultra-Fast Language Generation via Discrete Diffusion Divergence Instruct
The Fourteenth International Conference on Learning Representations, **(ICLR 2026)**

[5] Tiansheng Wen*, Yifei Wang*, Zequn Zeng, Zhong Peng, Yudi Su, **Xinyang Liu**, Bo Chen, Hongwei Liu, Stefanie Jegelka, and Chenyu You
Beyond Matryoshka: Revisiting Sparse Coding for Adaptive Representation
Oral Presentation [Top 1%]
Forty-Second International Conference on Machine Learning, **(ICML 2026)**

[6] **Xinyang Liu***, Hengrong Du*, Wei Deng, and Ruqi Zhang
Optimal Stochastic Trace Estimation in Generative Modeling
The 28th International Conference on Artificial Intelligence and Statistics, **(AISTATS 2025)**

[7] **Xinyang Liu***, Yilin He*, Bo Chen and Mingyuan Zhou
Advancing Graph Generation through Beta Diffusion
The Thirteenth International Conference on Learning Representations, **(ICLR 2025)**

[8] **Xinyang Liu***, Dongsheng Wang*, Bowei, Fang, Miaoge Li, Zhibin Duan, Yishi Xu, Bo Chen, and Mingyuan Zhou
Patch-Prompt Aligned Bayesian Prompt Tuning for Vision-Language Models
Proceedings of the 40th Conference on Uncertainty in Artificial Intelligence, **(UAI 2024)**

[9] Yishi Xu, Jianqiao Sun, Yudi Su, **Xinyang Liu**, Zhibin Duan, Bo Chen and, Mingyuan Zhou
Context-guided Embedding Adaptation for Effective Topic Modeling in Low-Resource Regimes
Thirty-seventh Conference on Neural Information Processing Systems, **(NeurIPS 2023)**

[10] Dongsheng Wang, Miaoge Li, **Xinyang Liu**, MingSheng Xu, Bo Chen and, Hanwang Zhang
Tuning Multi-mode Token-level Prompt Alignment across Modalities
Thirty-seventh Conference on Neural Information Processing Systems, **(NeurIPS 2023)**

[11] Miaoge Li*, Dongsheng Wang*, **Xinyang Liu**, Zequn Zeng, Ruiying Lu, Bo Chen, and Mingyuan Zhou
PatchCT: Aligning Patch Set and Label Set with Conditional Transport for Multi-Label Image Classification
The IEEE/CVF International Conference on Computer Vision, **(ICCV 2023)**

[12] Zhibin Duan*, **Xinyang Liu***, Yudi Su, Yishi Xu, Bo Chen, and Mingyuan Zhou
Bayesian Progressive Deep Topic Model with Knowledge Informed Textual Data Coarsening Process
In the 40th International Conference on Machine Learning, **(ICML 2023)**

Professional Services

- **Conference Reviewer:** NeurIPS (2024), ICML (2024, 2025, 2026), CVPR (2024, 2025), ICLR (2025), AISTATS (2025), AAAI (2025)

Teaching Experience

- **ELEMENTS OF REGRESSION ANALYSIS** Spring 2026
- **INTRO TO PROBABIL & STATISTICS** Fall 2025